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Spring fastener

Abstract

A spring fastener is provided. The spring fastener includes a spring and a flexible member having a shoulder, a first leg and a second leg. The first leg and the second leg are inserted through the spring along a longitudinal axis of the spring. The spring is retained to the flexible member by the shoulder and at least a portion of the first leg or the second leg. The first leg and the second leg have a retention feature to retain the spring fastener when the first leg and the second leg are inserted through an aperture in a component or a surface. A method of utilizing a spring fastener is also provided.

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Background/Summary

BACKGROUND

(1) Pushpins are commonly used for attaching a heat sink to a circuit board, so that the heat sink can help remove excess heat energy from a component. These are available in plastic or brass typically. Plastic pushpins suffer from long-term reliability, as a result of creep and aging. Brass pushpins, while not susceptible to creep and aging, do not have much springiness, and can damage circuit boards during installation. These situations are exacerbated on expensive circuit boards with large numbers of components and many pushpins. It is within this context that the embodiments arise.

SUMMARY

(2) In some embodiments, a spring fastener is provided. The spring fastener includes a spring and a flexible member having a shoulder, a first leg and a second leg. The first leg and the second leg are inserted through the spring along a longitudinal axis of the spring. The spring is retained to the flexible member by the shoulder and at least a portion of the first leg or the second leg. The first leg and the second leg have a retention feature to retain the spring fastener when the first leg and the second leg are inserted through an aperture in a component or a surface.

(3) In some embodiments, a spring fastener is provided. The spring fastener includes a spring and a shouldered member having at least two legs, formed by bending a flat piece that has a geometric center with a first leg and a second leg extending in opposed directions from the geometric center. At least one of the first leg or the second leg of the spring fastener includes a retention feature. The first leg and the second leg are arranged along a central, longitudinal axis of the spring with the spring retained by the shouldered member.

(4) In some embodiments, a method of using a spring fastener is provided. The method includes positioning, at an aperture of a component or a circuit board, pointed ends of two prongs of a pronged flexible member that retains a spring. The method includes pressing the pronged flexible member so that the two prongs enter the aperture and the pointed ends of the two prongs move towards each other. The method includes continuing to press the pronged flexible member so that the spring compresses as a result of contact with the component or the circuit board, and a retention feature of the two prongs secures the spring fastener to the component or the circuit board.

(5) Other aspects and advantages of the embodiments will become apparent from the following detailed description taken in conjunction with the accompanying drawings which illustrate, by way of example, the principles of the described embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The described embodiments and the advantages thereof may best be understood by reference to the following description taken in conjunction with the accompanying drawings. These drawings in no way limit any changes in form and detail that may be made to the described embodiments by one skilled in the art without departing from the spirit and scope of the described embodiments.

(2) FIG. 1 is a perspective view of a spring fastener in accordance with some embodiments.

(3) FIG. 2 is a perspective view of a shaped, flat piece that can be bent to form a flexible member

for the spring fastener of FIG. 1 in accordance with some embodiments.

(4) FIG. 3 is a top view of the shaped, flat piece of FIG. 2, showing two legs extending in opposed directions from a geometric center in accordance with some embodiments.

(5) FIG. 4A is a side projected view of a flexible member of the spring fastener of FIG. 1, formed from the shaped, flat piece of FIGS. 2 and 3 in accordance with some embodiments.

(6) FIG. 4B is a close-up view of a barbed prong of the flexible member of FIG. 4A in accordance with some embodiments.

(7) FIG. 5 is a side end projected view of the flexible member in accordance with some embodiments.

(8) FIG. 6 is a top projected view of the flexible member in accordance with some embodiments.

(9) FIG. 7A is a perspective view of the flexible member in accordance with some embodiments.

(10) FIG. 7B is a perspective view of a further embodiment of the flexible member of the spring fastener.

(11) FIG. 8 is a flow diagram of a method of using a spring fastener, which can be performed using the spring fastener of FIGS. 1-7 in accordance with some embodiments.

DETAILED DESCRIPTION

(12) Embodiments of a spring fastener disclosed herein have both spring properties and fastener properties, and can be used to fasten a heat sink or other component to a circuit board. Other uses may be devised. The spring fastener has a spring and a flexible member with two (or more, in some embodiments) legs, with a retention feature. In one embodiment, the retention feature has barbs on the ends of the legs, to both retain the spring and retain the fastener when inserted into an aperture, for example in a heat sink, a component or a circuit board.

(13) FIG. 1 is a perspective view of a spring fastener **102** in accordance with present embodiments. A flexible member **104** has a shoulder **120** and two legs **112** that retain a spring **106**. Further embodiments could have more than two legs. Tips **116** of the two legs have barbs **114** that act as a retention feature. The retention feature helps retain the spring **106** to the flexible member **104** and also helps to retain the spring fastener **102** to a component **118** such as a heat sink and/or to retain the spring fastener **102** and the component **118** to a further component (e.g., an integrated circuit package needing heat removal) or a circuit board (to which the component **118** is fastened). The spring **106** is a coil spring in this embodiment, with the legs **112** inserted through the coil spring along a central, longitudinal axis of the coil spring. Further types of springs, including flat springs, V springs, leaf springs and compound springs with more than one element are envisioned.

(14) An aperture **108** is defined in the head **110** of the spring fastener **102** along a bend portion of head **110**. This configuration places the aperture **108** near or proximate to the shoulder **120**, and not in the legs **112**. The dimensions of the aperture **108** control the flexibility of the head **110** of the spring fastener **102**, which in turn controls the flexibility of the spring fastener **102** in terms of pinching the legs **112** towards each other upon insertion of the tips **116** and legs **112** of the spring fastener into an aperture **122** in the component **118**. Also, the aperture **108** in the head **110** of the spring fastener **102** can be dimensioned to fit a tool used in pressing on the spring fastener **102** to insert the spring fastener **102** into a component **118** or circuit board. In some embodiments, the aperture **108** appears as a notch in the head **110**. A tool that has a projection that engages the aperture **108** or notch in the head **110** is readily devised.

(15) FIG. 2 is a perspective view of a shaped, flat piece that can be bent to form a flexible member **104** for the spring fastener of FIG. 1. The shaped, flat piece can be formed by stamping or otherwise cutting a flat sheet of metal, such as stainless steel, spring steel, other types of steel, brass, aluminum, etc., and bending along bend lines shown in FIGS. 1-4A and 5-7. These materials do not age or creep as does plastic, although an embodiment could be formed in plastic, fiberglass or other nonmetallic or composite material. As illustrated aperture **108** is formed in head **110** and the shaped flat piece can be bent along an axis of the aperture to form the flexible member **104**.

(16) FIG. 3 is a top view of the shaped, flat piece of FIG. 2, showing two legs **112** extending in

opposed directions from a geometric center. The geometric center forms the head **110** of the flexible member **104**, and has the aperture **108** in some embodiments. Edges along the boundary of the geometric center form the shoulder **120** of the flexible number **104**. In some embodiments, each of the legs **112** is at a nonzero angle **302** from a centerline symmetrical longitudinal axis of the geometric center of the shaped, flat piece. This nonzero angle **302** gives the tips **116** of the legs **112** a lateral offset or spacing in the shouldered member **104**, as seen in FIG. 4A. In some embodiments, the nonzero angle is about 3 degrees, however and nonzero angle may be utilized with the embodiments.

(17) FIG. 4A is a side projected view of a flexible member **104** of the spring fastener **102** of FIG. 1, formed from the shaped, flat piece of FIGS. 2 and 3. In this embodiment, a base **404** of one of the legs **112** overlaps a base **404** of the other leg **112**, where the legs **112** attach to the head **110** adjacent to the shoulder **120** of the flexible member **104**. The lateral offset or spacing of the tips **116** of the legs **112** is visible in the side projected view. Also visible is a further nonzero angle **402** between the legs **112**, arising from the nonzero angle **302** by which each leg is displaced from the centerline symmetrical longitudinal axis of the geometric center, as described above with reference to FIG. 3. As a result of the symmetrical folding of the geometric center to make the head **110** of the flexible member **104**, the further nonzero angle **402** between the legs **112** in the side projected view is equal to twice the nonzero angle **302** associated with each leg **112**. As illustrated in FIG. 4A, while the base **404** of one of the legs **112** overlaps the base **404** of the other leg **112** proximate to the head, tips **116** of the two legs **112** do not overlap in this embodiment.

(18) FIG. 4B is a close-up view of a barbed prong of the flexible member of FIG. 4A. In this embodiment, the leg **112** has a rounded tip **116**, with a barb **114** that projects out laterally to the side of the tip **116** of the leg **112**. The rounded tip allows the leg **112** to act as a prong, for easier insertion into an aperture of a component **118** or circuit board, in comparison to a squared off or blunt end of the leg. It should be appreciated that alternative tips, e.g., sharpened, etc. may be integrated with the embodiments. A beveled edge **404** is part of the tip **116** and the barb **114** of the leg **112**, and acts to guide the tip **116** of the leg **112** into the aperture. Beveling, in opposed directions, on the tips **116** of both of the legs **112** is arranged to push the two legs **112** closer together upon insertion of the tips **116** into an aperture during installation of the spring fastener **102**, as a result of the beveled edges **404** contacting edges of the aperture. With the flexible member **104** flexing so that the tips **116** of the legs **112** are thus squeezed closer together, the tips **116** and lower portion of the legs **112** can be pushed into the aperture in a component **118** or circuit board. Once the tips **116** are on the other side of the component or circuit board, a restoring force resulting from the flexing of the flexible member **104** acts to push apart the tips **116** of the flexible member **104**, so that the barbs **114** or other retention feature can move outward and press against that other side of the component or circuit board. This retains the spring fastener **102** to the component or circuit board. In further embodiments, the retention feature could be a hook, an L-bend, a latch, a ball, wings, a friction fit, a wedge, a slip on sheath, screw threads, threads for a nut, a cap, a bendable section, a twist tie, an aperture for insertion of a crossbar, or other feature that performs the functionality of retaining the spring fastener **102**.

(19) FIG. 5 is a side end projected view of the flexible member **104**. Bends, along the bend lines, can be seen shaping the head **110** and placing the legs **112** in plane parallel arrangement, with a spacing offset between the legs **112** in the side end projected view. One barb **114**, on the right, projects upwards out of the page in the drawing, the other barb **114** is not visible as it projects downward from the page in the drawing. Orientation of the barbs could readily be switched right to left in a variation, with a barb **114** on the left projecting upwards out of the page and a barb on the right projecting downward from the page. Head **110** is illustrated as having an elliptical shape in this side view. In other embodiments, alternative shapes, such as triangular, etc., may be integrated into the embodiments.

(20) FIG. 6 is a top projected view of the flexible member **104**. The aperture **108** and relationship

to remaining material of the head **110** of the flexible member **104** is visible. Removing more material, to enlarge the aperture **108**, would leave less material available for the head **116**, and decrease the amount of force needed to flex the head during displacement of the tips **116** and legs **112**. Conversely, shrinking the aperture **108** and leaving more material in the head **110** would increase the amount of force needed to flex the head during displacement of the tips **116** and legs **112**. The dimensions of the aperture **108** can thus be varied to control flexure of the flexible member **104**, for various materials. In some embodiments, the flexure of the flexible member **104** is tuned to be greater than flexure of a standard pushpin for a given amount of insertion force. It should be appreciated that this tuned embodiment protects against damage in circuit boards during assembly of the spring fastener **102** to a circuit board. As illustrated in FIG. **6**, the head **110** is bent along an axis of the aperture, although this is not meant to be limiting as the axis of the aperture may be offset from the bend line of the head, to one side of the middle in other embodiments. In addition to or instead of adjustment to the dimensions of the aperture **108**, adjustment of the geometry of the legs **112** and/or the angle of the prongs **114** affects the amount of force needed to flex the flexible member **104** during displacement of the tips **116** and legs **112**. For example, if the legs **112** have a wide root (at the shoulder **120**) and are narrower near the prongs **114**, the legs **112** flex more at the tips **116**, which helps distribute stress better. Also, by adjusting the prong ramp angle, the amount of friction between the tips **116** of the legs **112** and edges of a hole can be controlled. In some embodiments, the prong ramp angle is minimized.

(21) FIG. **7A** is a perspective view of the flexible member **104** without the spring **106**. Keeping in mind the shaping of the flexible member **104** from the bending of the flat piece shown in FIGS. **2** and **3**, and referring to FIGS. **1** and **7**, assembly of the spring fastener **102** is readily envisioned. The two legs **112** of the flexible member **104** are squeezed closer together, the tips **116** of the two legs **112** are inserted into the spring **106** from one end of the spring, and the coils of the spring **106** are pressed onto the legs **112**, or the legs **112** are pressed down into the coils of the spring **106**. Once the tips **116** of the legs **112** emerge from the other end of the spring **106**, the tips **116** and legs **112** move apart slightly, and the shoulder **120** and the legs **112** and/or the barbs **114** retain the spring **106** to the flexible member **104**.

(22) FIG. **7B** is a perspective view of a further embodiment of the flexible member **104** of the spring fastener **102**. The legs **112** overlap all the way to the tips **116**, i.e., for the entirety of the lengths of the legs **112** there is overlap from one leg **112** to the other leg **112**. This forces the legs **112** to slide with respect to each other during installation of the spring fastener **102**, which eliminates the legs **112** bumping into each other and also reduces insertion forces. Also visible is a vertical section **702** before the prong ramp angle **704**, so that the prong **114** has an increased contact area at that location when contacting the edge of the hole into which the spring fastener **102** is inserted during assembly. This reduces scratches in the component or printed circuit board into which the spring fastener **102** is inserted.

(23) FIG. **8** is a flow diagram of a method of utilizing a spring fastener, which can be performed using the spring fastener of FIGS. **1-7**. In an action **802**, pointed ends of prongs of a pronged spring fastener are positioned at an aperture of a component or a circuit board. In an action **804**, the pronged flexible member is pressed so that the prongs enter the aperture and the pointed ends of the prongs move towards each other. In an action **806**, the pronged flexible member continues to be pressed so that the spring compresses and a retention feature of the prongs secures the spring fastener to the component or circuit board. In a variation of the method, a tool is engaged to a notch or an aperture of the pronged flexible member, and used in pressing the pronged flexible member. It should be appreciated that the pronged spring fastener can secure a heat sink to a circuit board without use of an anchor bar as required by other types of fasteners.

(24) Detailed illustrative embodiments are disclosed herein. However, specific functional details disclosed herein are merely representative for purposes of describing embodiments. Embodiments may, however, be embodied in many alternate forms and should not be construed as limited to only

the embodiments set forth herein.

(25) It should be understood that although the terms first, second, etc. may be used herein to describe various steps or calculations, these steps or calculations should not be limited by these terms. These terms are only used to distinguish one step or calculation from another. For example, a first calculation could be termed a second calculation, and, similarly, a second step could be termed a first step, without departing from the scope of this disclosure. As used herein, the term “and/or” and the “/” symbol includes any and all combinations of one or more of the associated listed items.

(26) As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises”, “comprising”, “includes”, and/or “including”, when used herein, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. Therefore, the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting.

(27) It should also be noted that in some alternative implementations, the functions/acts noted may occur out of the order noted in the figures. For example, two figures shown in succession may in fact be executed substantially concurrently or may sometimes be executed in the reverse order, depending upon the functionality/acts involved.

(28) Although the method operations were described in a specific order, it should be understood that other operations may be performed in between described operations, described operations may be adjusted so that they occur at slightly different times or the described operations may be distributed in a system which allows the occurrence of the processing operations at various intervals associated with the processing.

(29) Various units, circuits, or other components may be described or claimed as “configured to” perform a task or tasks. In such contexts, the phrase “configured to” is used to connote structure by indicating that the units/circuits/components include structure (e.g., circuitry or mechanical features) that performs the task or tasks during operation. As such, the unit/circuit/component can be said to be configured to perform the task even when the specified unit/circuit/component is not currently operational (e.g., is not on). The units/circuits/components used with the “configured to” language include hardware—for example, circuits, memory storing program instructions executable to implement the operation, etc. Reciting that a unit/circuit/component is “configured to” perform one or more tasks is expressly intended not to invoke 35 U.S.C. 112, sixth paragraph, for that unit/circuit/component. Additionally, “configured to” can include generic structure (e.g., generic circuitry) that is manipulated by software and/or firmware (e.g., an FPGA or a general-purpose processor executing software) to operate in manner that is capable of performing the task(s) at issue. “Configured to” may also include adapting a manufacturing process (e.g., a semiconductor fabrication facility) to fabricate devices (e.g., integrated circuits or manufactured articles) that are adapted to implement or perform one or more tasks, or designing an article or apparatus to have certain features or capabilities.

(30) The foregoing description, for the purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain the principles of the embodiments and its practical applications, to thereby enable others skilled in the art to best utilize the embodiments and various modifications as may be suited to the particular use contemplated. Accordingly, the present embodiments are to be considered as illustrative and not restrictive, and the invention is not to be limited to the details given herein, but may be modified within the scope and equivalents of the appended claims.

Claims

1. A spring fastener, comprising: a spring; a shouldered member having at least .[two legs,].
.Iadd.a first leg and a second leg and being .Iaddend.formed by bending a flat piece.[.that.].Iadd., wherein the flat piece .Iaddend.has.Iadd.: .Iaddend. a geometric center with .[a.]. .Iadd.the .Iaddend.first leg and .[a.]. .Iadd.the .Iaddend.second leg extending in opposed directions from the geometric center, [.wherein the flat piece has.]. .Iadd.the first leg and the second leg each being offset by an angle from a symmetrical longitudinal axis of the geometric center; and .Iaddend. an aperture dimensioned to control flexure of the shouldered member; at least one of the first leg or the second leg having a retention feature; and the first leg and the second leg arranged .[along.].
.Iadd.relative to .Iaddend.a central, longitudinal axis of the spring with the spring retained by the shouldered member, wherein the flat piece is formed by stamping a flat sheet of metal to create the flat piece with the first leg and the second leg each offset by an angle from a symmetrical longitudinal axis of the geometric center of the flat piece.
2. The spring fastener of claim 1, wherein the retention feature includes a barb.
3. The spring fastener of claim 1, wherein the flat piece is cut and bent from a flat sheet of material.
4. The spring fastener of claim 1, wherein the aperture is proximate to a shoulder.
5. The spring fastener of claim 1, wherein the shouldered member and the spring are configured to secure a heat sink to a circuit board without use of an anchor bar.
6. The spring fastener of claim 1, wherein the first leg is at a nonzero angle with respect to the second leg.
7. The spring fastener of claim 1, wherein each of the first leg and the second leg is a beveled prong with a barb.
8. A spring fastener, comprising: a spring; a .[shouldered.]. .Iadd.flexible .Iaddend.member .Iadd.comprising a folded piece .Iaddend.having .Iadd.a head and .Iaddend.at least .[two.].
.Iadd.first and second .Iaddend.legs .[formed by bending a flat piece that has a geometric center with a first leg and a second leg extending in opposed directions from the geometric center,.].
.Iadd.extending from the head.Iaddend., wherein the .[flat piece.]. .Iadd.head .Iaddend.has an aperture dimensioned to control flexure of the .[shouldered.]. .Iadd.flexible .Iaddend.member; at least one of the first leg or the second leg having a retention feature; and the first leg and the second leg .[arranged along.]. .Iadd.disposed within a perimeter of the spring and extending from the head at a nonzero angle relative to .Iaddend.a central, longitudinal axis of the spring with the spring retained by the .[shouldered.]. .Iadd.flexible .Iaddend.member .Iadd.along the first and second legs.Iaddend., wherein: a shoulder of the .[shouldered.]. .Iadd.flexible .Iaddend.member secures a first end of the spring.[.; and an opposed second end of the spring is secured to the shouldered member by the retention feature or by the first leg and the second leg.].
9. The spring fastener of claim 8, wherein the at least one of the first leg or the second leg having a retention feature comprises: the first leg having a first barb; and the second leg having a second barb.
10. The spring fastener of claim 8, wherein the aperture is dimensioned to engage a tool to the spring fastener.
11. A spring fastener, comprising: a spring; a .[shouldered.]. .Iadd.flexible .Iaddend.member having .Iadd.a head and .Iaddend.at least .[two.]. .Iadd.first and second .Iaddend.legs .[formed by bending a flat piece that has a geometric center with a first leg and a second leg extending in opposed directions from the geometric center,.]. .Iadd.extending from the head,Iaddend.wherein the .[flat piece.]. .Iadd.head .Iaddend.has an aperture dimensioned to control flexure of the .
[shouldered.]. .Iadd.flexible .Iaddend.member.Iadd., and wherein the flexible member is formed by folding a piece about an axis lying on a vertical plane passing through a center of the aperture.Iaddend.; at least one of the first leg or the second leg having a retention feature; and the

first leg and the second leg .[.arranged along a central, longitudinal axis of the spring.].
.Iadd.disposed within a perimeter of the spring .Iaddend.with the spring retained by the .
[.shouldered.]. .Iadd.flexible .Iaddend.member, wherein: a base of the first leg .Iadd.at least
partially .Iaddend.overlaps a base of the second leg .[.in the shouldered member.]. .Iadd.proximate
the head of the flexible member when the flexible member is observed from a side view comprising
a viewing direction that is perpendicular to the vertical plane.Iaddend.; and a tip of the first leg is
spaced apart from a tip of the second leg .[.in the shouldered member.]. .Iadd.distal the head of the
flexible member when the flexible member is observed from the side view.Iaddend..
.Iadd.12. The spring fastener of claim 8, wherein respective distal ends of the first and second legs
partially coincide. .Iaddend.
.Iadd.13. The spring fastener of claim 8, wherein the first and second legs overlap along their entire
length. .Iaddend.
.Iadd.14. The spring fastener of claim 8, wherein the retention feature includes a barb configured to
secure the spring fastener to a circuit board. .Iaddend.
.Iadd.15. The spring fastener of claim 8, wherein each tip of the first leg and the second leg has a
barb..Iaddend.
.Iadd.16. The spring fastener of claim 8, wherein the folded piece comprises a flat piece cut from
sheet metal..Iaddend.
